

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 12A.1

INTRODUCTION TO ENTROPY

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	12: Entropy and Energetics
SUB-TOPIC:	12A.1: Introduction to Entropy
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	5 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 12A.1

Q1. Which of the following processes involves an increase in the entropy of the system? [WCH14/01/Jan23/Q1]

[1 Marks] **WCH14/01**

- A Freezing of water
- B Condensation of steam
- C Sublimation of dry ice
- D Precipitation of silver chloride

Q2. Explain why the standard entropy of gaseous carbon dioxide is greater than that of solid carbon dioxide. [WCH14/01/Oct22/Q3(a)]

[2 Marks] **WCH14/01**

Q3. Calculate the entropy change of the system for the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. Given standard molar entropies: $\text{N}_2(\text{g}) = 191.6$, $\text{H}_2(\text{g}) = 130.6$, $\text{NH}_3(\text{g}) = 192.3 \text{ J K}^{-1} \text{ mol}^{-1}$. [WCH14/01/Jan22/Q4(a)]

[2 Marks] **WCH14/01**

SECTION B: WORKED MARK SCHEME — 12A.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Which of the following processes involves an increase in the entropy of the system? [WCH14/01/Jan23/Q1] C	[1]
Q2	Q2: Explain why the standard entropy of gaseous carbon dioxide is greater than that of solid carbon dioxide. [WCH14/01/Oct22/Q3(a)] - Gases have more ways of arranging their particles / more disorder / more random motion than solids (1) - Gases have more microstates / energy is distributed over more energy levels (1)	[2]
Q3	Q3: Calculate the entropy change of the system for the reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. Given standard molar entropies: $\text{N}_2(\text{g}) = 191.6$, $\text{H}_2(\text{g}) = 130.6$, $\text{NH}_3(\text{g}) = 192.3 \text{ J K}^{-1} \text{ mol}^{-1}$. [WCH14/01/Jan22/Q4(a)] $\Delta S_{\text{system}} = (2 \times 192.3) - (191.6 + 3 \times 130.6) \text{ (1)} = 384.6 - 583.4 = -198.8 \text{ J K}^{-1} \text{ mol}^{-1}$ (1)	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 12A.1**FAQ #1: Why do students often get the sign of ΔS_{system} wrong for gas reactions?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why do students often get the sign of ΔS_{system} wrong for gas reactions?

MODEL ANSWER / EXAMINER EXPECTATION:

Students often forget to check the number of moles of gas on each side. If gas moles decrease, ΔS_{system} is negative. Always count the moles of gaseous reactants and products carefully.

FAQ #2: What is the common mistake in calculating standard entropy changes?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

What is the common mistake in calculating standard entropy changes?

MODEL ANSWER / EXAMINER EXPECTATION:

Candidates forget to multiply the standard entropy of each substance by its stoichiometric coefficient in the balanced equation.